Seema Rida

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Highly motivated Cognitive Science student at UC San Diego with a strong interest in Brain-Computer Interfaces (BCIs), neurotechnology, and computational neuroscience. Passionate about applying machine learning and signal processing to understand brain function and develop innovative neurotech solutions. As a quick learner, I am eager to innovate and contribute to a dynamic research environment, aiming to advance our understanding and technological integration in neuroscience.

EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO Expected Grad: June 2026
B.S. Cognitive Science spec/ Machine Learning and Neural Computation
Minor: Computer Science

- Clubs: Cognitive Science Student Association, Triton Prosthetics Society
- Relevant courses: Neural Signal Processing, Brain-Computer Interfaces, Supervised Machine Learning, Neuroanatomy and Physiology, Data Structures and Object-Oriented Design, Research Techniques and Methods

EXPERIENCE

Chiba Lab, La Jolla, CA October 2024-Present
Research Assistant (Part-Time)

- Assisting with canalysis and system development for the AI Classroom project, focused on real-time multimodal data capture.
- Configuring and troubleshooting network infrastructure for synchronized data collection.

ABWAAB, Amman, Jordan August 2024-December 2024
Data Science and Learning Intern (Part-Time)

- Conducted data analysis and feature engineering for a research project focused on student learning outcomes (under NDA).
- Worked with Abwaab's educational platform data to identify key behavioral and performance metrics related to student success.
- Collaborated with the data science team and learning team to explore predictive modeling strategies and refine feature sets for future applications.

THINKNEURO, Remote June 2024-August 2024
Co-Associate / Project Lead

- Conducted bibliometric analysis in RStudio, contributed to literature review on BCI therapies' impact on motor rehabilitation in stroke patients, and co-created a research poster presented at an e-symposium.
- Managed project tasks and schedules, ensuring team members met deadlines and coordinated effectively.

INTERNATIONAL ROBOTICS ACADEMY, Amman, Jordan June 2021-August 2023
Technical Instructor

- Led robotics classes for students aged 5 to 16, using EV3 and WeDo 2.0 kits for programming and design.
- Instructed students in Scratch programming, guiding them through interactive animation and gaming projects.
- Taught one-on-one sessions with students to effectively address individual learning needs and strengthen overall student understanding of foundational coding concepts.
- Assisted with administrative tasks such as managing payments, communicating with parents, and organizing schedules.

ABWAAB, Amman, Jordan July 2021-August 2021
UX Intern

- Designed a user-friendly profile page for the company website and production-live application using Figma.
- Collaborated with the software lead to design and implement an action plan for restructuring various website features, employing Figma mockups to refine design concepts for optimal user experience.

PROJECTS

Larval Zebrafish Neural Activity Analysis (*Python: pandas, scikit-learn, matplotlib, numpy)*

- Analyzed neural activity datasets in larval zebrafish to investigate processing patterns across brain regions.
- Performed data wrangling, PCA, and k-means clustering using Python (pandas, sklearn, matplotlib, numpy) to classify and visualize brain activity.

Zebrafish Muliscale Dynamics Analysis - In Progress *(Iterative Coarse Graining (ICG), Python)*

- Translated mathematical formulations from a neuroscience paper into a working implementation of the ICG algorithm.
- Applied the method to larval zebrafish neural activity data to uncover multiscale spatial structure across brain regions.

EEG-based attention analysis *(BrainFlow, Python, Signal Processing, EEG)*

- Designed and conducted EEG experiments to examine attentional fluctuations during task performance.
- Preprocessed EEG data (bandpass filtering, epoching), applied frequency domain analysis, and identified attention-related brainwave patterns.

Anticipatory Activity in a Songbird Vocal Motor Nucleus *(Spike train analysis, Python)*

- Analyzed spiking activity in a vocal motor nucleus of songbirds to investigate anticipatory patterns during song production.
- Compared firing rates pre- and post-song onset using time-locked spike data and statistical visualizations.

Automated Irrigation System *(Arduino, C++)*

- Developed a prototype using Arduino featuring a moisture sensor to detect when plants require watering, integrated with a mobile application to display real-time statistics. Implemented using C++.
- Nominated and then presented the prototype at Amman Design Week and the World Science Forum in Jordan.

- **Skills:** Python, Java, C, EEG data analysis, EEG data collection, machine learning, data analysis, Git, neural and vocal signal processing, network engineering, organization, mentoring, and teaching.